

Advanced JUnit Testing Exercises

Module 4 - Test Driven Development

JUnit 5 Advanced Testing Exercises

Exercise 1: Parameterized Tests

Source Code:

EvenChecker.java

```
package com.cognizant.advancedjunit;

public class EvenChecker {
    public boolean isEven(int number){
        return number % 2 == 0;
    }
}
```

EvenCheckerTest.java

```
package com.cognizant.advancedjunit;

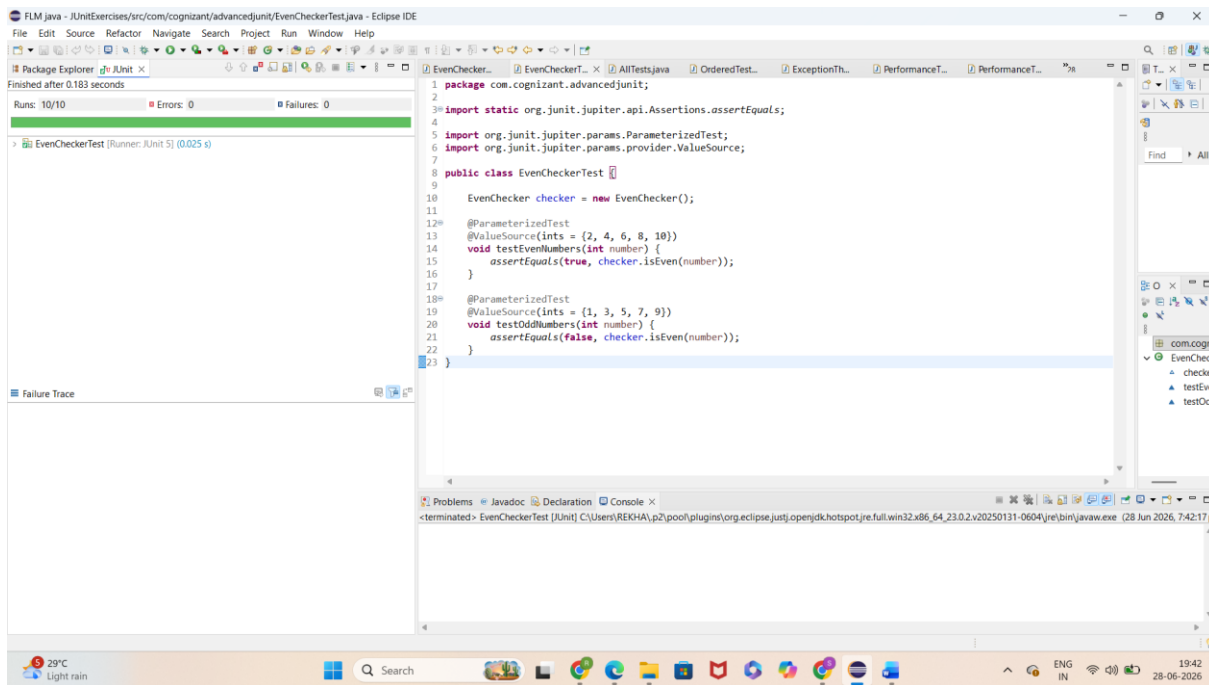
import static org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.params.ParameterizedTest;
import org.junit.jupiter.params.provider.ValueSource;

public class EvenCheckerTest {
    EvenChecker checker = new EvenChecker();

    @ParameterizedTest
    @ValueSource(ints={2,4,6,8,10})
    void testEvenNumbers(int number){
        assertEquals(true, checker.isEven(number));
    }

    @ParameterizedTest
    @ValueSource(ints={1,3,5,7,9})
    void testOddNumbers(int number){
        assertEquals(false, checker.isEven(number));
    }
}
```

Execution Output:



Exercise 2: Test Suites

Source Code:

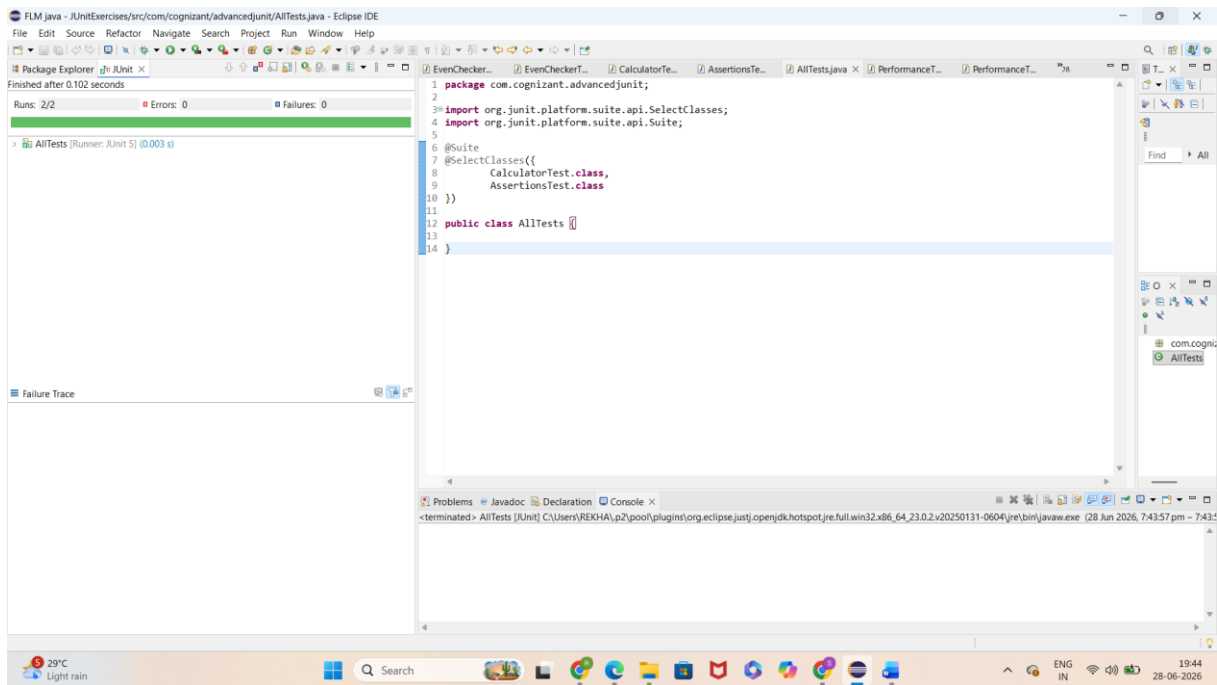
AllTests.java

```
package com.cognizant.advancedjunit;

import org.junit.platform.suite.api.SelectClasses;
import org.junit.platform.suite.api.Suite;

@Suite
@SelectClasses({CalculatorTest.class, AssertionsTest.class})
public class AllTests { }
```

Execution Output:



Exercise 3: Test Execution Order

Source Code:

OrderedTests.java

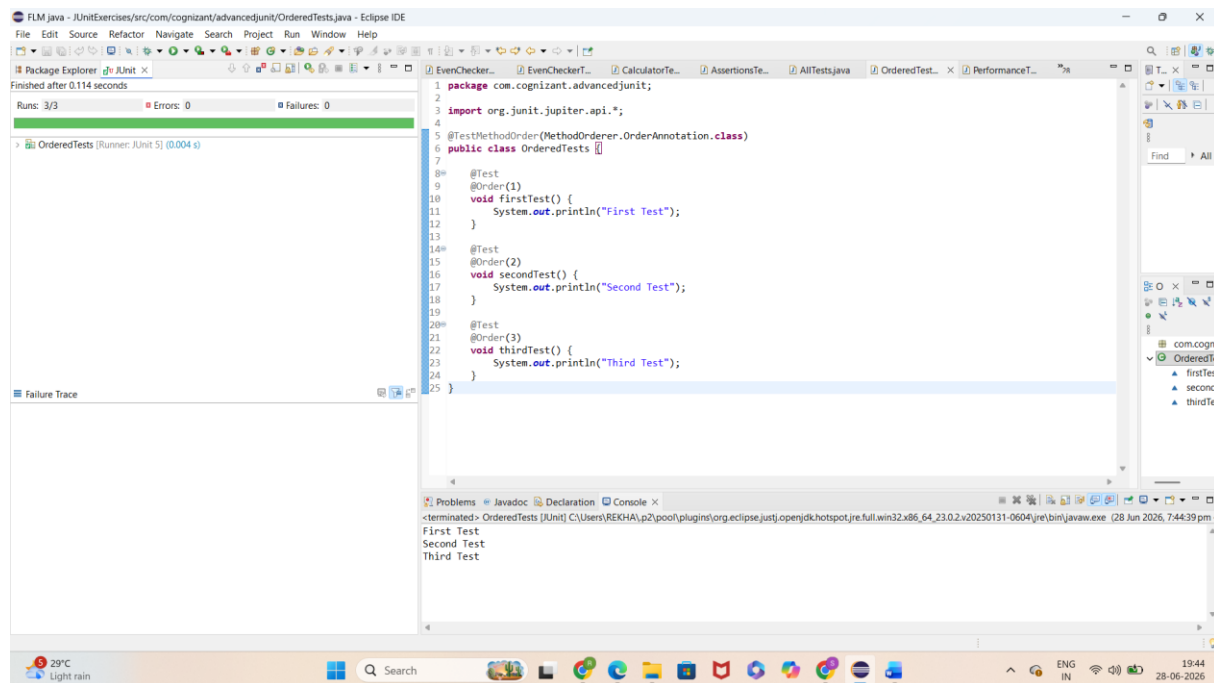
```
package com.cognizant.advancedjunit;

import org.junit.jupiter.api.*;

@TestMethodOrder(MethodOrderer.OrderAnnotation.class)
public class OrderedTests {

    @Test @Order(1) void firstTest(){ System.out.println("First Test"); }
    @Test @Order(2) void secondTest(){ System.out.println("Second Test"); }
    @Test @Order(3) void thirdTest(){ System.out.println("Third Test"); }
}
```

Execution Output:



Exercise 4: Exception Testing

Source Code:

ExceptionThrower.java

```
package com.cognizant.advancedjunit;

public class ExceptionThrower {
    public void throwException() {
        throw new ArithmeticException("Division by zero");
    }
}
```

ExceptionThrowerTest.java

```
package com.cognizant.advancedjunit;

import static org.junit.jupiter.api.Assertions.assertThrows;
import org.junit.jupiter.api.Test;

public class ExceptionThrowerTest {

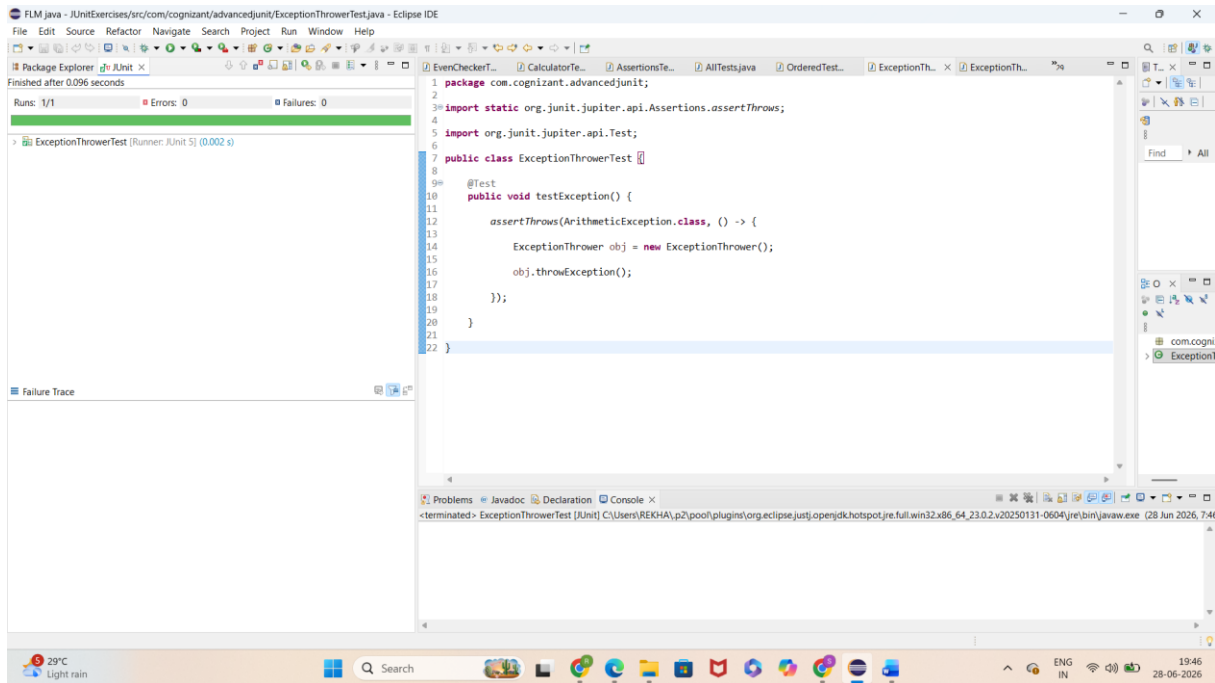
    @Test
    public void testException() {
        assertThrows(ArithmeticException.class, () -> {
            new ExceptionThrower().throwException();
        });
    }
}
```

```

    });
}
}

```

Execution Output:



Exercise 5: Timeout and Performance Testing

Source Code:

PerformanceTester.java

```

package com.cognizant.advancedjunit;

public class PerformanceTester {
    public void performTask() throws InterruptedException{
        Thread.sleep(1000);
    }
}

```

PerformanceTesterTest.java

```

package com.cognizant.advancedjunit;

import static org.junit.jupiter.api.Assertions.assertTimeout;
import java.time.Duration;
import org.junit.jupiter.api.Test;

```

```

public class PerformanceTesterTest {

    @Test
    public void timeoutTest(){
        PerformanceTester obj=new PerformanceTester();
        assertTimeout(Duration.ofSeconds(2), ()->{
            obj.performTask();
        });
    }
}

```

Execution Output:

